

1 **NeuroPilot: Longitudinal intravital neural imaging across scales and**
2 **modalities via agentic platform**

3

4 Abstract

5 Longitudinal intravital imaging is essential for understanding how neural populations
6 evolve across behavior, internal state, and time, yet translating large-scale recordings
7 into biological insight remains a fundamental barrier, as signals are obscured by noise,
8 motion-induced distortions, and heterogeneous imaging conditions.

9 Here we introduce NeuroPilot, a generalizable framework that establishes neural
10 imaging analysis as a closed-loop, data-conditioned process, integrating a multimodal
11 large language model with modular operators to adaptively optimize analysis from raw
12 data to biological inference.

13 Across diverse imaging modalities, biological preparations, and experimental regimes,
14 NeuroPilot enables robust and consistent neural signal recovery, supporting reliable
15 neuron detection and tracking in challenging settings including deep-brain imaging,
16 freely moving conditions, and longitudinal recordings spanning continuous 24-hour
17 timescales. This unified framework facilitates consistent analysis across heterogeneous
18 datasets, enabling behavioral decoding and revealing structured population dynamics
19 across conditions and time.

20 NeuroPilot establishes a scalable and autonomous paradigm for neural imaging analysis,
21 enabling systems-level interrogation of brain activity across expanded experimental
22 regimes.

Introduction

Deciphering how neural ensembles encode behavior and internal states across time remains one of the central challenges in neuroscience. Neural activity is inherently organized across coupled spatial and temporal scales, spanning rapid millisecond dynamics underlying sensory processing to long-term adaptations associated with learning, memory, pain, and circadian regulation. These dynamics are distributed across anatomically and functionally specialized brain regions, ranging from superficial cortical layers to deep neural circuits. Bridging these scales is essential for linking cellular-level activity to organism-level function, yet remains fundamentally difficult.

Recent advances in optical neural recording technologies have enabled in vivo monitoring of neuronal populations at single-cell resolution in behaving animals. These developments have opened unprecedented opportunities to study how distributed neural circuits give rise to behavior and internal states under naturalistic conditions. Tracking individual neuronal dynamics across time and experimental conditions provides insight into the stability and flexibility of neural representations. While data acquisition has rapidly scaled, the ability to extract reliable and interpretable signals from these recordings has not kept pace.

Capturing neural dynamics across extended timescales, particularly in freely behaving animals, remains a formidable technical and computational challenge. These longitudinal recordings are fundamentally constrained by the balance between signal integrity and phototoxicity, often resulting in photon-limited data with reduced signal-to-noise ratios. These limitations are further exacerbated when targeting deep brain regions, which serve as key substrates for higher-order processes including memory, decision-making, and emotional regulation. In these settings, imaging quality is often compromised by optical scattering, non-rigid tissue deformation, and motion-induced instability. Consequently, raw recordings frequently exhibit low signal fidelity and

signal mixing, posing a major barrier to the reliable extraction and interpretation of the neuronal activity underlying behavior and internal states.

Beyond data acquisition, a fundamental bottleneck lies in extracting biological insight from the rapidly growing volume of neural imaging data. Existing analysis pipelines typically rely on combinations of denoising, motion correction, segmentation, dimensionality reduction, and dynamical modeling. While these approaches have achieved substantial progress, they remain fragmented, require extensive manual tuning, and often fail to generalize across datasets, imaging modalities, and experimental conditions. Consequently, transforming large-scale neural recordings into robust, reproducible, and biologically meaningful representations remains a major unmet challenge in the field.

Here, we introduce NeuroPilot, an agentic framework for autonomous neural imaging analysis. By integrating multimodal large language models with modular computational tools, NeuroPilot achieves closed-loop, data-driven optimization of analysis workflows. Rather than relying on fixed pipelines, it dynamically adapts processing strategies to the specific quality and structure of each dataset. This adaptability ensures the robust restoration and extraction of neuronal signals even from highly challenging recordings, supported in part by a newly designed motion correction module tailored for complex, non-rigid imaging conditions. The framework reliably tracks individual neurons across extended timescales and generalizes seamlessly across diverse imaging modalities, brain regions, and behavioral paradigms. Ultimately, through iterative optimization and knowledge-guided decision-making, NeuroPilot unlocks the stable analysis of continuous 24-hour recordings, revealing coherent population dynamics, behavioral representations, and structured neural trajectories across circadian timescales. To facilitate accessibility and reproducibility, we provide

an interactive web interface that allows users to explore and apply NeuroPilot to their own data (<https://cebra.ai/> **TODO**).

Results

NeuroPilot , an agentic framework for autonomous and scalable neural imaging analysis

We developed NeuroPilot, an agentic computational framework for the autonomous and scalable analysis of neural imaging data. The framework is broadly compatible with one- to three-photon modalities and adapts to diverse experimental regimes, including longitudinal tracking, deep-tissue imaging, and multiplexed recordings (Fig. 1a). This development was motivated by the inherent limitations of raw neural recordings, where noise, motion artifacts, and signal mixing jointly degrade data interpretability and compromise both spike detection and spatial localization (Fig. 1b). By coordinating image restoration and motion correction via multimodal LLM-guided configuration, NeuroPilot achieves robust recovery of high-fidelity, biologically interpretable neural signals under heterogeneous imaging conditions.

At its core, NeuroPilot replaces conventional static pipelines with a closed-loop architecture, in which a multimodal large language model (LLM) dynamically orchestrates a suite of modular computational tools for denoising, motion correction, segmentation, and downstream quantitative analysis (Fig. 1c). Rather than executing a predefined sequence of steps, the system adaptively configures processing strategies based on input data characteristics and intermediate results. This adaptation also integrates information from a knowledge base comprising domain expertise, literature-derived heuristics, and execution history. Consequently, this design facilitates data-driven workflow optimization and ensures robustness across heterogeneous imaging regimes.

The computational backbone of NeuroPilot is an iterative refinement loop that tightly couples self-supervised denoising and motion correction under multimodal LLM supervision (Fig. 1d; Supplementary Fig. 1). Within each iteration, denoising first enhances signal quality to facilitate accurate motion estimation, after which a non-rigid registration module predicts dense deformation fields to align the imaging stack. This iterative feedback drives the progressive refinement of both signal fidelity and spatial alignment. Throughout the process, the LLM continuously evaluates intermediate outputs and adaptively reconfigures processing strategies, such as tuning parameters or re-initializing modules when necessary, to ensure robust convergence across heterogeneous imaging conditions. Upon completing this optimization, the LLM orchestrates downstream analyses, including segmentation and activity extraction, and generates a traceable report summarizing both the processing workflow and key characteristics of the recording data.

NeuroPilot is designed to scale into a hierarchical framework for the analysis of large-scale imaging datasets. Within this architecture, individual imaging stacks are first processed independently and subsequently integrated through cross-stack registration, footprint alignment, and the joint analysis of neuronal activity. This design extends the framework from local stack-wise restoration to unified analysis across multiple datasets, enabling trace comparison at extended temporal and experimental scales. Such a formulation is particularly well-suited for longitudinal imaging across days, ultra-long recordings segmented into temporal blocks, and datasets collected under diverse experimental conditions. By combining local restoration with global integration, NeuroPilot preserves the fidelity of each individual stack while ensuring stable and scalable analysis across large-scale recording datasets.

Transformer-based non-rigid motion correction for neural imaging

To seamlessly integrate with the self-supervised denoising module, we developed a transformer-based motion correction module that performs dense, non-rigid alignment via optical flow estimation. At its core, a pyramid-based local correlation registration network (PyLoReg-Net) estimates optical flow through a coarse-to-fine strategy ([Supplementary Figs. 2-3](#)). The network extracts multi-scale feature pyramids and computes local correlation cost volumes to iteratively refine deformation fields, enabling the accurate modeling of both large-scale displacements and fine local distortions.

PyLoReg-Net is trained using a hybrid objective that combines optical flow supervision with an image restoration loss, ensuring that motion estimates improve spatial alignment while preserving signal fidelity. The training dataset comprises both experimentally acquired recordings and clean synthetic images ([Supplementary table 1](#)). To facilitate the supervised learning of dense motion fields, we generate training pairs by applying simulated affine, elastic, and non-linear deformations to these images. The warped images serve as inputs, with the known deformation fields providing ground-truth targets. This paradigm ensures accurate motion estimation across diverse conditions and robust generalization to challenging in vivo scenarios. During inference, the trained model is incorporated into the NeuroPilot pipeline, where the predicted deformation fields iteratively align image stacks within the closed-loop framework.

Validation of NeuroPilot using paired corrupted and ground-truth recordings

To evaluate the performance of NeuroPilot, we devised a paired acquisition strategy that enables direct comparison between degraded recordings and matched high-quality references. We implemented a custom two-channel two-photon imaging system, in which fluorescence emission from the same field of view is partitioned by a 1:9 beam splitter into low- and high-signal detection paths, yielding paired datasets with distinct degradation levels ([Supplementary Fig. 4a](#)). The raw high-signal recordings were

150 motion-corrected to establish a reference dataset, while the corresponding low-signal
151 recordings were processed by NeuroPilot ([Supplementary Fig. 4b](#)). This design provides
152 a controlled and quantitative benchmark for assessing restoration fidelity at both
153 structural and functional levels.

154 We first evaluated NeuroPilot on paired datasets to assess its ability to restore
155 neuronal structure and activity under controlled degradation. In representative fields of
156 view comprising mixed populations of somata and dendrites, corrupted recordings
157 exhibited poorly defined structures and distorted spatiotemporal kymographs.
158 Following NeuroPilot processing, the restored images achieved enhanced structural
159 continuity and contrast, closely mirroring the paired reference recordings ([Fig. 2a-c](#)).
160 Magnified views further highlighted the accurate recovery of fine neuritic features and
161 temporally coherent signal fluctuations. To determine whether these structural gains
162 translate into functional accuracy, we compared calcium traces extracted from the
163 corrupted, restored, and reference datasets. NeuroPilot-derived signals faithfully
164 matched the reference temporal profiles, preserving both event timing and amplitude,
165 whereas traces from the corrupted recordings suffered from attenuated amplitudes and
166 elevated noise ([Fig. 2d,e](#)). Quantitatively, NeuroPilot restoration yielded an
167 approximately two-fold increase in Pearson correlation and a three-fold improvement in
168 signal-to-noise ratio (SNR) ([Fig. 2f,g](#)). Furthermore, the estimated displacement fields
169 revealed complex, non-uniform motion within the corrupted data, underscoring the
170 necessity of motion-aware restoration ([Fig. 2h](#)).

171 We further evaluated NeuroPilot on recordings featuring fine dendritic structures,
172 which are inherently susceptible to degradation from low contrast and motion artifacts.
173 NeuroPilot substantially enhanced the clarity of dendritic morphology and recovered
174 coherent spatiotemporal patterns that closely mirrored the reference data ([Fig. 2i-k](#)).
175 Accordingly, the extracted activity traces demonstrated strong concordance with the

reference signals. Quantitative metrics confirmed these gains, with Pearson correlation increasing from approximately **0.25 to 0.80** and signal-to-noise ratio (SNR) rising from **1.5 to 6.0** following restoration (Fig. 2l-o). Furthermore, motion analysis revealed substantial deformation along both spatial axes in the corrupted recordings, reinforcing the necessity of jointly addressing noise and non-rigid motion at the subcellular level (Fig. 2p-s).

To assess population-level performance, we evaluated additional paired datasets. NeuroPilot markedly improved soma detectability, identifying 150 neurons compared to just 35 in the corrupted data, yielding a population size comparable to the reference condition (118 neurons) (Supplementary Fig. 5b). Population-wide analysis revealed that NeuroPilot restoration substantially increased temporal correlation with reference traces (**from ~0.2 to ~0.85**) and enhanced the signal-to-background ratio (SBR, from **~1.0 to ~1.5**) (Supplementary Fig. 5c). Finally, representative traces demonstrated the faithful preservation of neuronal dynamics and robust suppression of background noise (Supplementary Fig. 5d).

Stable identification of spinal cord neurons across diverse stimuli and longitudinal timescales

In vivo spinal cord calcium imaging is inherently limited by low signal-to-noise ratios and severe non-rigid motion. This tissue deformation is particularly exacerbated in freely moving subjects and under sensory stimulation, where the spinal cord is mechanically coupled to active musculature. By employing NeuroPilot, we achieve robust identification of individual neurons across diverse behaviors, stimulus paradigms, and extended recording timescales. Utilizing a miniature two-photon spinal cord imaging preparation equipped with a custom groove window (Fig. 3a), we performed repeated imaging of the same field of view (FOV) across diverse stimulus conditions and longitudinal sessions. This approach allowed for cross-stimulus alignment of

neuronal populations within a shared FOV across multiple behavioral and sensory contexts (Fig. 3b). Simultaneously, longitudinal recordings under consistent experimental paradigms or pain models facilitated the re-identification of the same neurons over extended timescales (Fig. 3c).

Across all four conditions, raw recordings were severely degraded by noise, motion artifacts, and condition-dependent tissue deformation, resulting in fragmented spatial structures and discontinuous temporal signals. NeuroPilot successfully restored high-resolution morphology and spatial coherence while significantly enhancing temporal stability (Fig. 3d-g). Kymograph analysis further demonstrated near-complete suppression of spatiotemporal jitter and the recovery of continuous signal trajectories. Consequently, individual neurons were reliably tracked across stimuli, with 21 neurons consistently matched across all four paradigms (Fig. 3h).

Functionally, NeuroPilot enhanced the temporal continuity and signal fidelity of extracted calcium traces, yielding reduced noise and clearer activity profiles compared to raw recordings (Supplementary Fig. 7a,b). These improvements facilitated more robust estimation of neuronal activity and pairwise correlations. Population activity heatmaps derived from NeuroPilot-processed data revealed condition-dependent temporal patterns in $\Delta F/F$ signals across behavioral states (Fig. 3i-l). Correspondingly, pairwise correlation matrices exhibited distinct population-level relationships across stimuli (Fig. 3m), indicating stimulus-specific variations in population activity patterns.

We extended this approach to longitudinal spinal cord imaging in a spared nerve injury (SNI) model over a 28-day period. Despite progressive fluctuations in imaging conditions and substantial tissue deformation, NeuroPilot facilitated the consistent identification of 25 neurons across five recording sessions (Days 1, 5, 14, 21, and 28) (Fig. 3n-s). While raw recordings exhibited pronounced structural degradation and

temporal instability across sessions, NeuroPilot restored high-resolution neuronal morphology and spatiotemporal consistency, as evidenced by enhanced cellular clarity and stabilized kymograph patterns (Supplementary Fig. 8a,b). Spatial footprint alignment and displacement analysis further demonstrated that cross-day positional shifts were effectively compensated, enabling stable registration to a common reference frame (Fig. 3t). Quantitative analysis confirmed that NeuroPilot substantially suppresses frame-to-frame motion along both spatial axes, reducing the large-amplitude drift observed in raw recordings to near-zero levels (Supplementary Fig. 8c,d).

This spatial stabilization directly underpins improved longitudinal tracking and functional consistency. As shown in Supplementary Fig. 9, NeuroPilot preserves neuronal identities and spatial footprints across days while yielding calcium traces with superior temporal fidelity compared to raw data. Population correlation matrices further reveal a combination of conserved and evolving ensemble patterns across sessions. Quantitative analysis confirms that neurons can be reliably identified across flexible session subsets, demonstrating robust registration across arbitrary temporal intervals (Fig. 3v-x). Collectively, these results establish NeuroPilot as a robust framework for stable neuron identification across both dynamic behavioral states and extended timescales, providing a foundation for analyzing spinal cord neural dynamics over weeks-long periods.

Generalization across imaging modalities and spatial scales of NeuroPilot

We utilized NeuroPilot for head-mounted three-photon calcium imaging targeting deep brain regions (Fig. 4a). In volumetric reconstructions spanning cortical and hippocampal depths (~1,000 μm), raw recordings exhibit pronounced signal attenuation and noise accumulation, which obscure neuronal structures. NeuroPilot restores depth-resolved spatial organization across cortical layers and subcortical regions, including L2/3, L4/5, L6, the corpus callosum, and hippocampal CA1 (Fig. 4b,

Supplementary Fig. 10). Representative fields of view at multiple depths (495 μm , 670 μm , and 908 μm) demonstrate consistent recovery of neuronal morphology throughout the axial range (Fig. 4c). Correspondingly, segmentation maps reveal well-defined neuronal spatial footprints following restoration (Fig. 4e). Calcium activity traces extracted from these neurons exhibit enhanced temporal fidelity, characterized by reduced noise and clearer transient dynamics across all depths (Fig. 4d). During long-duration recordings (~ 100 min), NeuroPilot maintains both spatial and temporal consistency. Maximum intensity projections and kymographs reveal enhanced structural clarity and signal continuity (Fig. 4f), while motion trajectories indicate substantial suppression of tissue displacement (Fig. 4g). Population activity maps and representative traces confirm stable signal extraction over extended periods (Fig. 4h).

We next examined We next evaluated the generalization of NeuroPilot to one-photon miniscope imaging, which offers a broader field of view (FOV) relative to three-photon recordings. In a representative Day 1 session (Fig. 4i-l), NeuroPilot restored spatial organization, resolved distinct neuronal footprints, and suppressed both motion-induced artifacts and fixed-pattern noise. These enhancements were consistently maintained across longitudinal recordings (Days 2 and 3), ensuring stable spatial footprints and high-fidelity temporal signals (Supplementary Fig. 11). Collectively, these results demonstrate the robust performance of NeuroPilot in one-photon miniscope imaging across multiple days. Similar efficacy was observed in mesoscale two-photon datasets (Supplementary Fig. 10), where NeuroPilot enhanced contrast and recovered fine structural details across millimeter-scale (~ 6 mm) fields of view.

Robust recovery across deep-brain and diverse biological imaging targets

We further demonstrated the robust capabilities of NeuroPilot in the demanding regime of deep-brain calcium imaging, successfully resolving diverse biological targets spanning neuronal, vascular, and subcellular signals. In the exceptionally challenging

context of deep-brain two-photon miniscope imaging of the lateral entorhinal cortex (LEC, at an extreme depth of ~ 4.3 mm; [Fig. 5a](#)) and hippocampal CA1 (~ 1.2 mm; [Fig. 5m](#)), where raw recordings suffered from severe signal attenuation and scattering, NeuroPilot effectively restored spatial organization. Maximum intensity projections following restoration enabled the clear delineation of neuronal morphology across these regions ([Fig. 5b,n](#)). Notably, segmentation analysis revealed a dramatic expansion in detectable neuronal populations; in CA1, the number of identified neurons increased nearly eightfold, from 52 to 414 ([Fig. 5o](#)). Extracted calcium traces exhibited significantly enhanced temporal fidelity, characterized by suppressed baseline noise and sharpened transient dynamics ([Fig. 5d,p](#)). Quantitative metrics further confirmed these improvements: motion trajectory analysis demonstrated the effective elimination of tissue displacement, reducing micrometer-scale fluctuations to near-zero levels ([Fig. 5e-g](#)). Fourier ring correlation (FRC) analysis revealed a substantial gain in spatial resolution, improving from ~ 6 μm to ~ 2 μm in LEC and from ~ 18 μm to ~ 2 μm in CA1 ([Fig. 5i,q](#)). Simultaneously, signal-to-background ratios (SBR) were consistently elevated ([Fig. 5j,r](#)), reflecting markedly improved contrast and signal separability at these extreme depths. Finally, population-level analyses indicated that the temporal organization of neuronal activity was preserved, as evidenced by similarity matrices and low-dimensional manifolds ([Fig. 5k,l,s,t](#)).

Beyond neuronal somata, NeuroPilot exhibits robust generalization across diverse biological structures. In dual-channel recordings, both functional and structural signals are concurrently preserved, maintaining precise spatial registration and consistent temporal profiles ([Supplementary Fig. 12](#)). In vascular imaging, NeuroPilot recovers continuous tubular architectures and their associated dynamics, confirming the effective preservation of non-neuronal organization ([Supplementary Fig. 13](#)). Furthermore, in neuronal mitochondrial calcium imaging, the framework restores intricate network-like structures and enhances local signal continuity. This facilitates the

clear delineation of mitochondrial morphology and branching patterns while faithfully maintaining temporal activity fidelity ([Supplementary Fig. 13](#)).

NeuroPilot supports continuous 24-hour imaging of behavioral-state-dependent population dynamics

We utilized NeuroPilot for continuous 24-hour calcium imaging of medial prefrontal cortex (mPFC) neuronal populations in freely behaving mice, facilitating simultaneous analysis of neural activity and behavioral dynamics across circadian timescales ([Fig. 6a,b](#)). Behavioral recordings were synchronized with calcium imaging data, encompassing a broad repertoire of states including sleep, locomotion, grooming, and feeding ([Supplementary Fig. 14](#)). This approach yields a longitudinal perspective of neuronal activity spanning the full sleep-wake cycle. NeuroPilot yielded significant enhancements in signal quality throughout the recording duration. Compared to raw data, restored images exhibited superior contrast and clearer delineation of neuronal morphology ([Fig. 6c,d](#)). Corresponding kymographs revealed enhanced temporal continuity and mitigated background fluctuations, while extracted calcium traces displayed improved signal clarity with preserved transient dynamics ([Fig. 6e](#)). These improvements were consistently observed across diverse behavioral states, with processed traces exhibiting suppressed noise while maintaining high temporal fidelity ([Supplementary Fig. 15](#)).

These enhancements ensured robust long-term stability. Population-averaged $\Delta F/F$ remained stable throughout the 24-hour recording, with NeuroPilot maintaining values at approximately 0.25, nearly double the 0.12 observed in raw data, without significant decay over time ([Fig. 6f](#)). Similarly, the signal-to-noise ratio (SNR) was markedly improved and stabilized at ~ 13 , whereas raw recordings exhibited a progressive decline from 7 to 4 ([Fig. 6g](#)). Motion analysis further demonstrated the effective suppression of cumulative tissue displacement. While raw recordings showed

progressive drift exceeding 15 μm along both spatial axes, NeuroPilot-stabilized trajectories remained tightly constrained, with residual displacement consistently below 1 μm (Fig. 6h; Supplementary Fig. 16). This synergistic signal and spatial stability facilitated consistent frame-to-frame alignment, enabling reliable longitudinal tracking of individual neurons over the full 24-hour duration (Supplementary Fig. 17).

At the single-neuron level, NeuroPilot preserves calcium transient kinetics and temporal structure over prolonged durations, facilitating consistent comparisons of neuronal activity across behavioral states (Fig. 6i). At the population level, the framework enables the extraction of large neuronal ensembles characterized by well-defined spatial footprints (Fig. 6j). Analysis of behavior-aligned mean $\Delta F/F$ revealed that a substantial fraction of neurons exhibited preferential activation during REM sleep relative to other behavioral states (Fig. 6k). Mapping these activity profiles to their anatomical coordinates demonstrated that neurons with elevated REM-associated activity are spatially distributed throughout the field of view (Fig. 6l). Similar trends were observed across multiple animals (Supplementary Fig. 18), confirming the reproducibility of these activity patterns.

Decoding and dynamical organization of mPFC population activity across the behavioral spectrum

To characterize its temporal organization, we projected population activity into a low-dimensional space. Although the separation between behavioral states was subtle, the trajectories exhibited a gradual progression throughout the session and converged toward their initial ZT0 coordinates at ZT24, consistent with circadian periodicity (Fig. 7a-c; Supplementary Fig. 19). Principal component loadings were broadly distributed across the neuronal population, indicating that this low-dimensional structure was not dominated by a small subset of cells (Fig. 7d).

We investigated the extent to which population activity supports behavioral decoding. Utilizing a multinomial logistic regression model, we predicted behavioral states across multiple levels of granularity, including binary sleep-wake classification, coarse behavioral categories, and fine-grained ethogram annotations (Fig. 7e-g). Sleep-wake decoding achieved high accuracy, with diagonal classification rates of ~0.8-0.9 (Fig. 7e), whereas coarse behavioral categories exhibited moderate separability, with diagonal values typically ranging from 0.5 to 0.7 (Fig. 7f). At the fine-grained level, decoding performance varied across states (Fig. 7g,i), reflecting heterogeneous discriminability across behavioral categories. Confusion matrices revealed that prediction errors were non-uniformly distributed, occurring more frequently between behaviorally related states, such as locomotion and ingestive behaviors, or microarousals and sleep (Fig. 7g). These misclassification patterns suggest partial overlap in population activity representations across behaviors (Fig. 7h). Hierarchical clustering further demonstrated that certain behaviors aggregate based on activity similarity without forming discrete, strictly separated clusters (Fig. 7j). These trends were consistently observed across animals (Supplementary Fig. 20).

To investigate population activity during sleep-state transitions, we extracted data segments corresponding to NREM-to-REM transitions; the resulting heatmaps illustrate the temporal evolution of neuronal dynamics (Fig. 7k-n; Supplementary Fig. 21). To further characterize these trajectories, we employed recurrent switching linear dynamical system (rSLDS) modeling of sleep-related activity. This analysis identified discrete dynamical regimes defined by distinct vector fields that govern the evolution of neural trajectories (Fig. 7k, l-o; Supplementary Fig. 22). Notably, REM trajectories exhibited outward expansion in state space, whereas NREM trajectories remained more confined within local attractor regions. Consistently, the Euclidean distance between instantaneous states and inferred attractors was significantly higher in REM than in NREM (Fig. 7p), reflecting state-dependent differences in the dispersion of population

activity across sleep states. Consistent with these observations, pairwise correlations varied over time and across behavioral conditions ([Supplementary Fig. 23](#)). At the population level, the average correlation exhibited gradual temporal fluctuations aligned with behavioral annotations ([Supplementary Fig. 24](#)).

Discussion

Neuroimaging technologies have advanced rapidly, enabling large-scale, long-duration recordings across diverse brain regions and behavioral contexts. However, transforming these data into reliable, interpretable representations remains a fundamental bottleneck, as signal quality is frequently compromised by noise, motion artifacts, and heterogeneous imaging conditions. Here, we present NeuroPilot, an agentic framework that integrates multimodal large language models (MLLMs) with modular computational tools to facilitate adaptive, closed-loop optimization of neuroimaging analysis. In contrast to rigid pipelines, NeuroPilot dynamically configures and refines processing strategies based on dataset characteristics, ensuring consistent signal recovery across diverse experimental regimes.

Central to NeuroPilot is the coordination of multiple processing modules within an iterative, data-driven framework. This design diverges from conventional sequential workflows, where modules operate independently with static parameters. Instead, NeuroPilot facilitates the interdependent refinement of intermediate outputs, allowing improvements in one stage to inform subsequent processing steps. By leveraging LLM-guided orchestration, the system adaptively selects and tunes analysis strategies according to data quality, imaging modality, and experimental context. Such flexibility is critical in neuroimaging, where inherent variability across preparations and recording conditions often limits the generalizability of fixed pipelines.

NeuroPilot consistently enhances signal fidelity across spatial and temporal dimensions. These improvements transcend individual frame restoration to ensure

spatiotemporal consistency, enabling stable tracking of neuronal signals under adverse conditions, including pronounced motion, low signal-to-noise ratios, and long-term recordings. Notably, NeuroPilot supports reliable neuron identification across longitudinal imaging sessions and consistent alignment across diverse behavioral contexts. Collectively, these results demonstrate that NeuroPilot facilitates the extraction of stable, comparable signals from datasets that are otherwise refractory to reliable analysis.

These capabilities generalize across imaging modalities and biological scales. NeuroPilot performs robustly in one-, two-, and three-photon calcium imaging, spanning large fields of view, deep-brain recordings, and volumetric imaging regimes. Extending beyond somatic imaging, the framework also recovers diverse biological targets, such as vascular structures and subcellular signals like mitochondrial calcium dynamics. In each case, NeuroPilot enhances structural continuity and temporal consistency without requiring modality-specific reconfiguration. This broad applicability suggests that the framework captures fundamental principles of signal degradation and recovery rather than overfitting to specific experimental configurations.

Beyond improving raw signal quality, NeuroPilot enables the extraction of population-level representations that remain stable over protracted timescales. In continuous 24-hour recordings, the framework maintains consistent signal amplitude, signal-to-noise ratio, and spatial alignment, supporting the tracking of neuronal activity across full circadian cycles. These stabilized datasets facilitate downstream analyses that would otherwise be computationally or analytically prohibitive, including behavioral decoding, population trajectory analysis, and the characterization of state-dependent neural dynamics. While the resulting representations remain primarily descriptive, they provide a coherent view of population activity across time, behavior, and experimental conditions.

The agentic architecture of NeuroPilot points to several promising avenues for future development. By decoupling analysis from rigid pipelines and enabling adaptive, data-driven optimization, this framework could be extended toward autonomous workflows that dynamically recalibrate processing strategies in response to evolving data quality and experimental conditions. Such capabilities are particularly valuable for long-duration or unconstrained behavioral experiments, where recording conditions inherently fluctuate. In principle, these approaches could be integrated into online or closed-loop imaging systems, facilitating real-time feedback between data acquisition and analysis.

Beyond automation, the ability of NeuroPilot to generate stable, comparable representations across modalities and timescales provides a foundation for large-scale data integration. As neuroimaging studies increasingly rely on multi-session, multi-condition, and multi-subject datasets, consistent signal extraction is essential for meaningful cross-dataset comparisons. Building on this framework, future work could enable systematic alignment and joint analysis, facilitating investigations of population dynamics across diverse individuals, behavioral contexts, and longitudinal timescales. More broadly, these representations may support the integration of heterogeneous datasets into unified analytical spaces, fostering new paradigms in comparative and population-level neuroscience.

Methods

Image restoration and denoising module

To restore neural imaging data degraded by noise and low signal-to-noise ratio, we developed a self-supervised image restoration module based on a three-dimensional U-Net architecture, trained directly on raw recordings without the need for clean ground-truth data.

Network architecture The denoising model was implemented using a 3D U-Net architecture operating on spatiotemporal image volumes. The network consists of a multi-level encoder–decoder structure with skip connections that preserve high-resolution spatial information while capturing hierarchical features. The encoder path is composed of sequential convolutional blocks followed by downsampling operations, progressively increasing the number of feature channels. Each block consists of two consecutive 3D convolution layers with non-linear activation (ReLU) and optional normalization. The decoder path performs symmetric upsampling and feature fusion by concatenating encoder features with upsampled representations, followed by convolutional refinement. A final $1 \times 1 \times 1$ convolution maps the features to the restored output volume. All convolutions operate in three dimensions (time $\times$ height $\times$ width), enabling the model to jointly exploit spatial and temporal redundancy in calcium imaging data.

Self-supervised training strategy The model was trained in a self-supervised manner using only noisy recordings, without requiring paired clean targets. To construct training pairs, input patches were generated from raw data using structured masking strategies that exploit redundancy in space and time. Specifically, multiple sampling modes were implemented:

Spatial splitting (XY mode): input and target patches were constructed by partitioning pixels along spatial dimensions (e.g., interleaved rows or columns).

Temporal splitting (T mode): input and target patches were constructed from alternating frames along the temporal dimension.

Pixel dropout (Noise2Void-style): a subset of pixels was randomly masked in the input, while the original signal served as the training target.

These strategies enable the network to learn denoising by predicting withheld information from surrounding spatial and temporal context. The model was trained using the Adam optimizer. The loss function combined L1 and L2 reconstruction losses.

Inference and volumetric reconstruction During inference, full image stacks were processed using a patch-based strategy to accommodate large volumes. Input volumes were divided into overlapping 3D patches, and each patch was independently processed by the trained network. The denoised patches were then merged to reconstruct the full volume, with overlap regions ensuring smooth transitions and reducing boundary artifacts. To ensure compatibility with the network architecture, input patches were padded to match required dimensions and cropped back to original size after processing.

Transformer-based motion correction (PyLoReg-Net)

To correct non-rigid motion in neural imaging data, we developed a transformer-enhanced optical flow model, termed PyLoReg-Net, which combines pyramid-based local correlation with global feature interaction for accurate motion estimation across multiple spatial scales.

Multi-scale feature extraction Input image pairs were first processed using a convolutional encoder to extract hierarchical feature representations at multiple spatial resolutions. Features were organized as a pyramid from coarse to fine scales. At each scale, feature maps from the reference frame and the moving frame were used for motion estimation, allowing the model to capture both large-scale displacements and fine local deformations.

Transformer-based global feature interaction To enhance long-range correspondence, we introduced a lightweight transformer module that performs global interaction between feature maps of the two frames prior to motion estimation. At selected low-resolution scales, feature maps were flattened into token sequences and augmented with 2D sinusoidal positional embeddings. Multi-head self-attention and cross-attention were then applied symmetrically to both feature sets, enabling information exchange between distant spatial regions. This transformer-based

interaction improves the robustness of correspondence estimation, particularly under large motion or low signal-to-noise conditions.

Local correlation and cost volume construction Following feature interaction, motion was estimated using a PWC-style local matching mechanism. For each spatial location, a cost volume was constructed by computing normalized correlations between features within a local neighborhood in the second frame. The resulting cost volume encodes matching confidence across candidate displacements and serves as the input to a convolutional refinement network.

Flow estimation Dense optical flow was estimated from the cost volume using a convolutional refinement network within a coarse-to-fine framework. At the coarsest scale, an initial flow field was predicted, and at subsequent finer scales, features from the moving frame were warped using the upsampled flow from the previous level. Residual flow fields were iteratively estimated and accumulated to progressively refine the motion estimate, enabling accurate modeling of both large displacements and subtle local deformations. Flow fields estimated at lower resolutions were then progressively upsampled to higher resolutions using learnable modules, including transposed convolution and pixel shuffle, with scale-adaptive upsampling factors to ensure consistency between feature resolution and motion estimation.

Synthetic and experimental training data for PyLoReg-Net

To train and evaluate the motion correction model (PyLoReg-Net), we used a combination of large-scale synthetic datasets and experimentally acquired two-photon calcium imaging recordings.

Synthetic data generation using NAOMi Synthetic training data were generated using the NAOMi simulation framework, which provides a physics-based pipeline for realistic two-photon microscopy data synthesis. The simulation pipeline consisted of five sequential modules: (1) Neural volume generation. Three-dimensional neural volumes were generated, including neuronal somata, dendrites, axons,

vasculature, and background structures, with realistic spatial distributions. (2) Neural activity simulation. Temporal activity traces were generated using stochastic spiking models, producing neuron-specific calcium dynamics with configurable firing rates and temporal resolution. (3) Optical propagation modeling. Optical effects were simulated using point spread function (PSF) modeling based on system parameters such as numerical aperture (NA), excitation power, and imaging depth. (4) Scanning simulation. The neural volume and activity were combined through a simulated raster scanning process to generate time-lapse image sequences mimicking two-photon imaging acquisition. (5) Noise modeling. Realistic noise sources were incorporated during scanning, including photon shot noise and background fluctuations, to match the statistical properties of experimental data.

Using this pipeline, we generated large-scale synthetic datasets with varying imaging depths, neuronal densities, firing rates, and optical conditions. In our implementation, volumes of size up to $500 \times 500 \times 60$ voxels were simulated, with temporal sequences of several thousand frames per recording. Multiple simulation runs were performed with varying parameters to increase diversity and improve model generalization.

Experimental datasets In addition to synthetic data, we incorporated multiple publicly available two-photon calcium imaging datasets for training and validation. These datasets span diverse brain regions, imaging conditions, and signal-to-noise regimes. A summary of the datasets used is provided in [Supplementary Table 1](#), including: Neurofinder benchmark datasets and CRCNS two-photon datasets (pvc and ssc series). These datasets collectively comprise hundreds of imaging stacks and over one million frames, covering a wide range of imaging conditions.

Neuron segmentation and trace extraction

Neuron segmentation and activity extraction were performed on NeuroPilot-processed image stacks using a neurite-aware pipeline based on Suite2p. For each

recording, maximum intensity (MAX) and standard deviation (STD) projections were computed to capture complementary structural and activity-related features, and segmentation was conducted with parameters optimized for elongated and overlapping structures. Neuropil correction and baseline normalization were applied to extract fluorescence activity traces. To improve segmentation quality, an intensity-based filtering step was applied using normalized MAX and STD images. For each ROI, mean intensity within its spatial footprint was computed, and ROIs that were simultaneously dim in both representations were removed, preserving structures that are either structurally prominent or functionally active. For downstream analysis, ROI masks were further refined by retaining the largest connected component and ranking ROIs based on the maximum amplitude of their fluorescence traces. A subset of high-confidence ROIs was selected, and masks were optionally polished using morphological operations to improve spatial consistency. The resulting ROI masks and $\Delta F/F$ traces were used for all subsequent analyses.

Cross-session and longitudinal alignment

To enable consistent tracking of neural activity across sessions and over extended timescales, we performed cross-session alignment using a two-stage pipeline combining image restoration and optical flow-based registration. For each imaging session, raw image stacks were first processed using NeuroPilot to remove noise and correct intra-session motion artifacts, yielding denoised and motion-stabilized recordings. From each processed stack, standard deviation (STD) projections and maximum intensity projections (MIP) were computed, capturing complementary functional and structural features, respectively.

Dense non-rigid transformations between sessions were estimated using PyLoReg-Net applied to pairs of STD and MIP images. These projections provide robust representations for cross-session correspondence, allowing accurate estimation of spatial deformation fields even under substantial variability in signal intensity and

activity patterns. The resulting optical flow fields encode pixel-wise displacements between sessions. The estimated deformation fields were subsequently applied to the full spatiotemporal image stacks by warping each frame to a common reference coordinate system. This strategy enables efficient and consistent alignment of entire recordings while preserving temporal dynamics. The resulting dataset is fully aligned across sessions, enabling direct comparison of neuronal activity over time and supporting downstream analyses such as longitudinal neuron tracking and population-level dynamics.

Imaging datasets and experimental conditions

One-photon miniscope imaging One-photon calcium imaging was performed in the dorsal CA1 (dCA1) region of the hippocampus using a miniscope system with GRIN lens implantation. GCaMP6f was expressed via AAV injection (AP -2.1 mm, ML 2.0 mm, DV -1.65 mm). A stacked GRIN lens configuration was used, consisting of a 1.8 -mm-diameter, 4.7 -mm-long lens (Edmund Optics) implanted above CA1 (lens center: DV -1.25 mm), enabling imaging of the pyramidal layer. Calcium imaging was performed at 23 Hz for 15 min per session across three sessions on separate days. Imaging data were acquired at a resolution of 1000×1000 pixels and cropped to the GRIN lens field of view (~ 1.8 mm diameter, 704×704 pixels).

Sciatic nerve injury (SNI) model To induce neuropathic pain, the sciatic nerve injury (SNI) model was established as previously described. Mice were first subjected to baseline imaging (defined as Day 1 in this study). Immediately following imaging, animals underwent SNI surgery. Briefly, mice were deeply anesthetized with 1.5 – 3% isoflurane delivered in 100% oxygen. After sterilization with 70% ethanol, animals were placed in a prone position and the limbs were secured to an anatomical plate. A dorsal longitudinal skin incision was made spanning vertebral levels T11 to L1 to expose the underlying musculature. The biceps femoris muscle was bluntly separated using forceps to reveal the sciatic nerve and its three terminal branches: the sural,

common peroneal, and tibial nerves. Neuropathic injury was induced by tightly ligating the common peroneal and tibial nerves using 5-0 silk sutures (RockTech), followed by transection distal to the ligation site and removal of approximately 4 mm of the distal nerve stump. The sural nerve was left intact. The incision site was then closed and sutured using 5-0 silk sutures.

Population-level analysis and low-dimensional representation

To characterize the structure and temporal evolution of neural population activity, we performed population-level analyses using low-dimensional representations derived from principal component analysis (PCA).

Preprocessing and population representation Neuronal fluorescence signals were converted to $\Delta F/F$ and organized into a time-by-neuron matrix. Behavioral labels and time indices were aligned to each time point. To ensure computational efficiency while preserving global population structure, a subset of time points was randomly sampled for fitting the dimensionality reduction model, and the learned transformation was subsequently applied to the full dataset.

Low-dimensional representation using PCA Neural population activity was represented in a low-dimensional space using principal component analysis. To improve computational stability, an initial dimensionality reduction step was performed, projecting the data onto an intermediate subspace (PCA-pre). The data were then projected onto the top 10 principal components (PCA10), which served as the final population-level representation. This PCA10 representation captures the dominant modes of population activity and was used for all downstream analyses, including visualization, clustering, and trajectory analysis.

Behavioral decoding and representational similarity analysis

To characterize the association between neural population activity and behavioral states, we combined supervised behavioral decoding with representational similarity analysis (RSA) leveraging confusion structure and population statistics.

Data preprocessing and feature construction Neuronal activity traces were converted to $\Delta F/F$ traces and structured as a time-by-neuron matrix. To maintain computational tractability and ensure balanced sampling across extended recordings, the data were temporally subsampled to a fixed resolution. Each time point was associated with a behavioral label derived from synchronized annotations, with categorical states encoded as integer indices. Prior to decoding, neural features were z-score normalized using statistics derived exclusively from the training set; these identical parameters were then applied to the test set to preclude data leakage.

Train-test splitting with temporal structure preservation To evaluate decoding performance while accounting for temporal dependencies, we implemented a block-wise train-test splitting strategy. The time series was partitioned into contiguous blocks, a subset of which was allocated to the test set. To ensure evaluative robustness, we required all behavioral classes to be represented in both partitions. If this criterion remained unmet after iterative resampling, a stratified random split was utilized as a fallback. This approach effectively mitigates temporal leakage while maintaining comprehensive class representation across data splits.

Behavioral decoding models Behavioral states were decoded from neural population activity using a linear support vector machine (LinearSVM). The classifier was trained to predict behavioral labels based on instantaneous population activity vectors. A linear decoder was selected to provide an interpretable measure of population-level separability while mitigating the risk of overfitting in high-dimensional neural data. Class imbalance was addressed via class-weighting where appropriate.

Decoding performance was assessed on held-out test data using multiple metrics, including overall accuracy, balanced accuracy, and the macro-averaged F1 score.

Confusion matrix analysis and representational similarity To characterize the structure of decoding errors and identify relationships between behavioral states, we analyzed confusion matrices derived from model predictions. The matrices were normalized on a per-class basis to estimate the conditional probability of predicted states given the ground truth. These normalized matrices capture the representational similarity between neural population activity patterns across different behaviors. To formally quantify this similarity, we constructed a symmetric similarity matrix:

$$S = \frac{1}{2}(C + C^T)$$

where C denotes the normalized confusion matrix. Hierarchical clustering was then applied to this similarity matrix to identify groups of behaviors with similar neural representations.

Directed confusion structure and asymmetry analysis To further characterize directional relationships between behavioral representations, we computed an asymmetry matrix:

$$A = C - C^T$$

which quantifies directional biases in misclassification between behavioral states. We additionally constructed directed confusion graphs, where nodes represent behavioral states and edges correspond to dominant misclassification pathways. Edge weights were derived from the largest off-diagonal elements of the confusion matrix. Chord diagrams were used to visualize these directed relationships in a compact form.

Dynamical systems modeling (rSLDS)

To characterize the temporal organization and dynamical structure of neuronal population activity, we employed a recurrent switching linear dynamical system (rSLDS), which approximates nonlinear dynamics as a latent mixture of locally linear regimes. Neuronal activity traces ($\Delta F/F$) were first projected into a lower-dimensional

observation space using Principal Component Analysis (PCA). Specifically, the population activity matrix (time $\times$ neurons) was decomposed into its top principal components (typically 10), preserving dominant variance while attenuating noise and redundancy. These reduced observations were then modeled via the *ssm* framework. Within this rSLDS architecture, neural population trajectories transition through a set of discrete dynamical states, each governed by an underlying linear dynamical system. At each time point t , the latent state $x_t \in R^D$ evolves according to:

$$x_{t+1} = A_{z_t} x_t + b_{z_t} + \epsilon_t$$

Where $x_{t+1} \in \{1, \dots, K\}$ denotes the discrete state, A_{z_t} and b_{z_t} define the linear dynamics for that state, and ϵ_t is Gaussian noise. The discrete state transitions are governed by a recurrent transition model that depends on the continuous latent state, enabling context-dependent switching between dynamical regimes. Model fitting was performed using variational inference (Laplace-EM or BBVI), with latent dimensionality set to 2 for visualization and interpretation of population trajectories. Initialization was performed using PCA-based projections to ensure stable convergence.

After training, we extracted both the continuous latent trajectories x_t and the most likely discrete state sequence z_t . The learned system matrices A_k and offsets b_k for each discrete state were directly retrieved from the trained model. To visualize the underlying dynamics, we computed the velocity field associated with each dynamical regime. Specifically, the local flow at state x was approximated as:

$$v(x) = (A_k - I)x + b_k$$

which represents the instantaneous displacement of the latent state under the learned dynamics. Velocity fields were evaluated over a grid spanning the latent space and visualized using vector field plots. Latent trajectories were overlaid onto these fields, allowing direct comparison between observed neural dynamics and the inferred dynamical structure. Trajectories were further color-coded either by behavioral state or by normalized time to reveal state-dependent and temporal evolution patterns.

To summarize global dynamics, we additionally computed an occupancy-weighted mixture flow field, in which the velocity fields of individual modes were combined according to their empirical usage frequencies:

$$v_{mix}(x) = \sum_k \pi_k v_k(x)$$

Where π_k denotes the fraction of time spent in each discrete state.

Data availability

All relevant data that support the findings of this study are available from the corresponding authors upon reasonable request.

Code availability

Our Brainfound can be found at <https://github.com/gingerbread000/Brainfound>.

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Author contributions

Q.D., X.L., and Y.G. conceived the Brainfound project and revised the manuscript. G.Z. and Z.G. implemented the Brainfound framework, trained the multimodality model, completed the fine-tuning of downstream tasks, organized the experimental results, and composed the manuscript. C. D. collected data and established the BrainCT-3M and BrainMRI-7M datasets. J.L. completed the saliency visualization of Brainfound attention. T.W., Y.G., and Y.C. established the BrainInstru-1M instruction dataset. L. W. collected 5T MRI brain data. Y. L., Y. L., Q. C., K. F., and L. W. completed the human-machine evaluation of automatic report generation. Q. C., K. F., and Y. L. Q. C., L. W., and Y. L. completed the answering of BrainMCQ.

749 **Competing financial interests**

750 The authors declare no competing financial interests.

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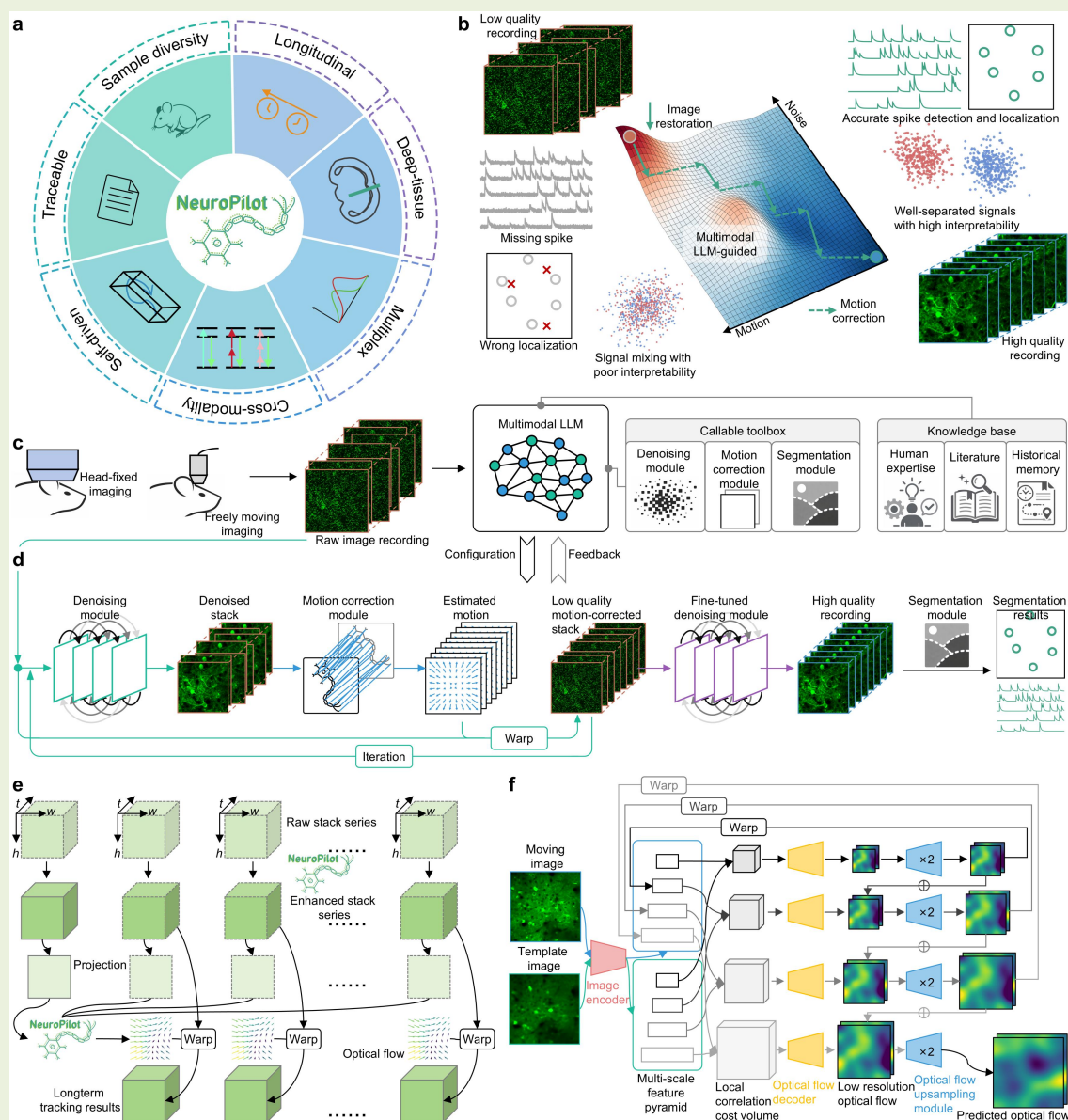


Fig. 1 | NeuroPilot for multimodal neural imaging analysis.

a, Overview of the NeuroPilot. As an agentic platform, NeuroPilot enables robust and scalable analysis of complex neuronal recordings. NeuroPilot handles a spectrum of advanced experimental demands, ranging from diverse samples and longitudinal tracking to deep-tissue imaging and multiplexing. It demonstrates broad compatibility across single- to three-photon modalities. Driven by autonomous configuration and traceable processing, NeuroPilot provides a integrated workflow for large-scale neural data evaluation.

b, Motivation for the NeuroPilot. Raw recordings suffer from noise and motion artifacts that degrade signal interpretability, leading to missing spikes, incorrect localization, and signal mixing. NeuroPilot employs multimodal LLM-guided decision making to

orchestrate image restoration and motion correction. This coordinated processing yields high-quality recordings, enabling accurate spike detection, precise spatial localization, and highly discriminable neural signals.

c, Architecture of NeuroPilot. Raw image recordings acquired from different experimental paradigms are ingested by a multimodal large language model (LLM). The multimodal LLM dynamically orchestrates a callable toolbox comprising denoising, motion correction, and segmentation modules, while leveraging a knowledge base encompassing human expertise, literature knowledge, and historical memory to drive pipeline optimization.

d, Autonomous iterative processing pipeline. Guided by a multimodal LLM, NeuroPilot initializes and continuously refines the workflow through active result feedback. The sequence begins with primary denoising and motion estimation. By employing an iterative loop that encompasses spatial warping alongside a fine-tuned denoising step, the system progressively upgrades intermediate stacks into high-fidelity recordings. These optimized datasets are routed to the segmentation module to extract precise neuronal activity signals.

e, Hierarchical cross-session processing with NeuroPilot. Raw stacks from different sessions or conditions are first enhanced, followed by projection-based optical flow estimation. The resulting deformation fields are used to warp each stack into a common reference frame, enabling consistent alignment and long-term tracking across days and stimuli.

f, Coarse-to-fine registration in PyLoReg-Net for motion correction. Moving and template images are encoded into shared multi-scale feature pyramids. Optical flow is estimated progressively from low ($H/16 \times W/16$) to high ($H/2 \times W/2$) resolution. At each scale, a local correlation cost volume between template and warped moving features is used to predict residual flow, which is added to the upsampled flow from the previous level to refine the dense deformation field.

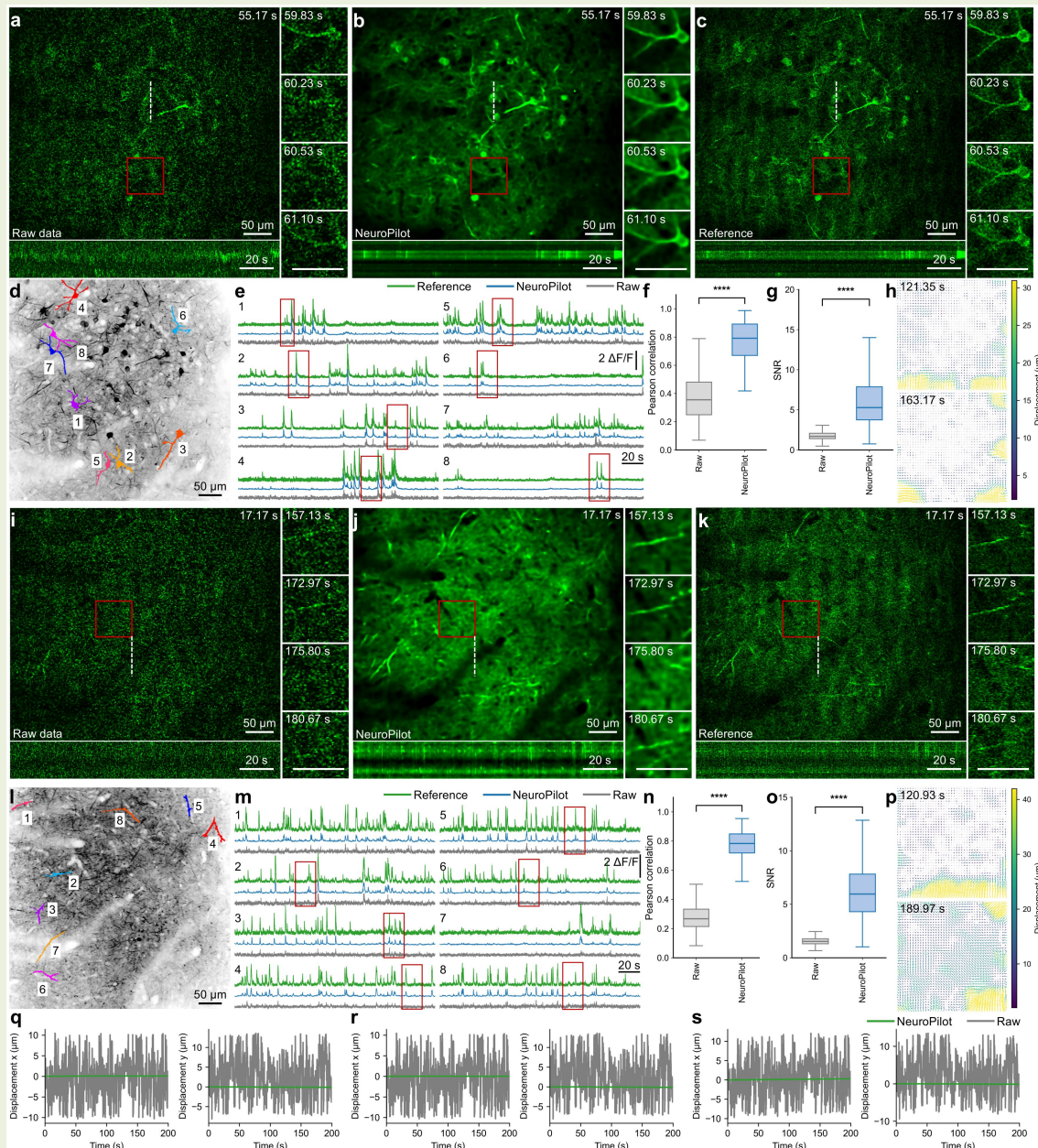


Fig. 2 | NeuroPilot robustly restores neural signals from corrupted recordings.

a-c, Example field of view (FOV) containing a mixed population of neuronal somata and dendritic processes. Comparison of the raw low-SNR recording **a**, the corresponding NeuroPilot-restored result **b**, and the paired high-quality ground-truth reference **c**. Red boxes indicate zoomed regions in the corresponding insets. Scale bars, 50 μm . Temporal kymographs in the bottom panels, generated along the white dashed line, highlight the effective removal of motion artifacts. Time scale bars, 20 s.

d, Representative reference image highlighting eight color-coded neurons selected for subsequent activity comparison. Scale bar, 50 μm .

e, Extracted calcium activity traces from the eight representative neurons identified in **d**. Signals recovered by NeuroPilot (blue) reproduce those from the high-quality ground-truth reference (green), rescuing the severe noise present in the raw recordings (gray). Red boxes highlight specific calcium spikes accurately restored by NeuroPilot. Scale bars, 2 $\Delta F/F$ vertically and 20 s horizontally.

f, Box plots show the Pearson correlation coefficients of raw and NeuroPilot-processed calcium traces compared to the ground-truth reference. $P < 0.0001$ (calculated using two-sided paired t-test; $n = \text{TODO 78}$ neurons). For the box plots, the center line indicates the median, box limits represent the 25th and 75th percentiles, and whiskers extend to the minimum and maximum values.

g, Box plots demonstrate a substantial SNR improvement for extracted neuronal signals after NeuroPilot restoration compared to the raw recordings. $P < 0.0001$ (two-sided paired t-test; $n = \text{TODO 78}$ neurons). Box plot definitions are the same as in **f**.

h, Representative motion displacement maps at two distinct timepoints (121.35 s and 163.17 s) illustrate the substantial, non-uniform motion artifacts present in the raw recording. Arrows indicate the direction of local tissue displacement, with magnitudes color-coded according to the scale bar.

i-k, NeuroPilot accurately recovers fine neurite morphology. A second example FOV dominated by dendritic structures is shown, comparing the raw low-SNR data **i**, NeuroPilot-restored result **j**, and ground-truth reference **k**. Display conventions, including zoomed insets, temporal kymographs, and scale bars, are the same as in a-c.

l, Representative reference image highlighting eight color-coded dendritic segments selected for subsequent activity comparison. Scale bar, 50 μm .

m, Extracted calcium activity traces from the eight representative dendritic segments identified in **l**. Signal representations, highlighted regions (red boxes), and scale bars are the same as in **e**.

n, Pearson correlation coefficients demonstrating improved signal similarity for the extracted dendritic segments. $P < 0.0001$ (two-sided paired t-test; $n = \text{[TODO] 83}$ dendritic segments). Box plot conventions match those described in **f**.

o, NeuroPilot substantially enhances the SNR of extracted dendritic signals. $P < 0.0001$ (two-sided paired t-test; $n = \text{[TODO] 83}$ dendritic segments). Box plot definitions are as in **f**.

p, Representative motion displacement maps for the dendritic FOV at two distinct timepoints (120.93 s and 189.97 s). Vector field annotations and color-coding conventions are identical to those in **h**.

q-s, Temporal motion trajectories along the x- and y-axes. The graphs display the continuous positional fluctuations in the raw recordings (gray) and the resulting trajectories maintained near zero after NeuroPilot processing (green). The data in **q**, **r**, and **s** are extracted from the recordings shown in **a**, **i**, and [Supplementary Fig. 4](#), respectively.

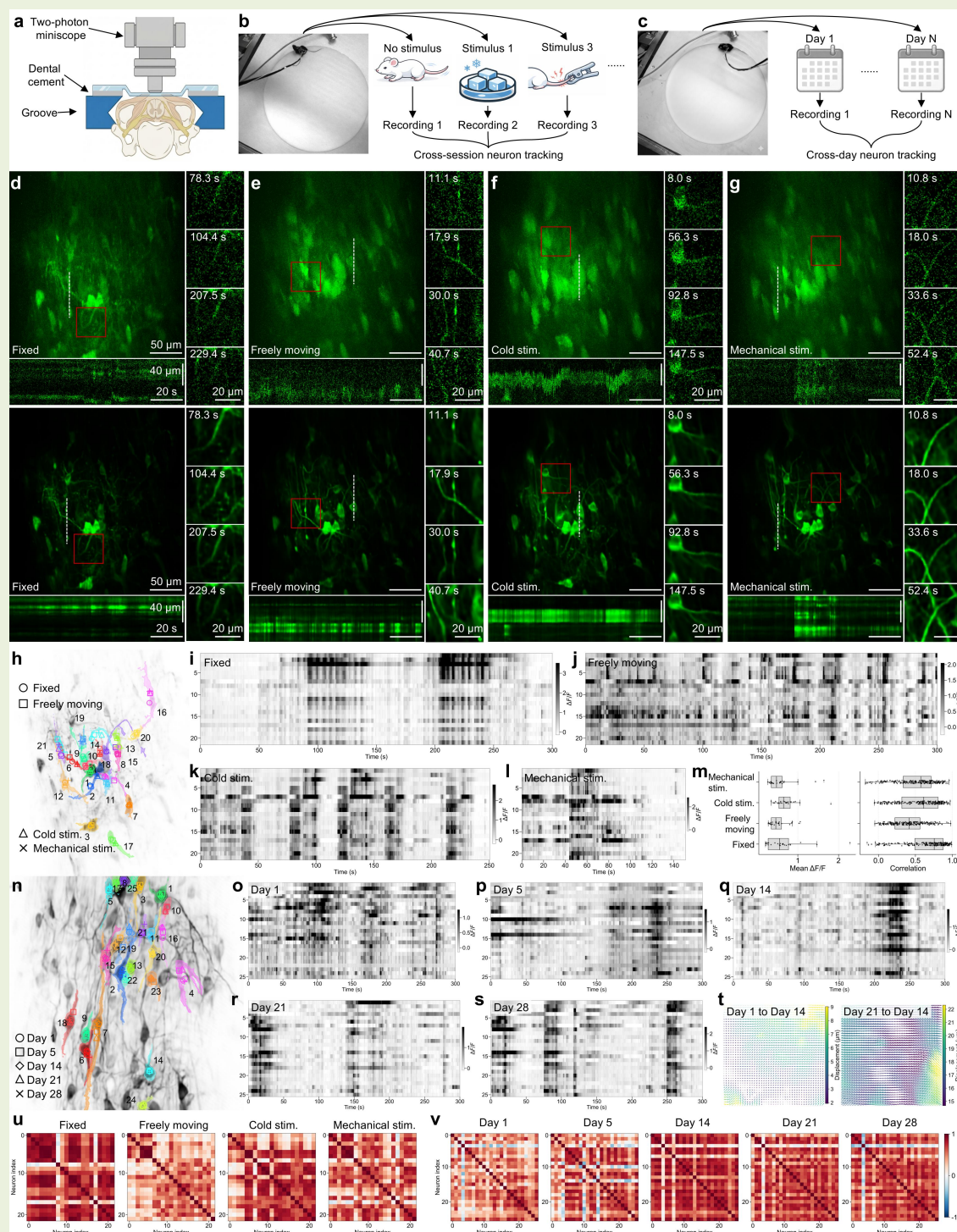


Fig. 3 | NeuroPilot achieves longitudinal and cross-stimulus tracking of spinal cord neurons.

a, Schematic of the in vivo spinal cord imaging setup. A two-photon miniscope is mounted above a custom groove window and secured with dental cement.

b, Cross-session tracking across distinct sensory stimuli within a single day.

c, Longitudinal tracking of the same neuronal population across multiple days.

d-g, Example spinal cord imaging FOVs across four experimental conditions: **spinal column-fixed** **d**, freely moving **e**, cold stimulation **f**, and **nociceptive mechanical stimulation via tail pinch** **g**. The top and bottom rows display the maximum intensity projections of the raw and NeuroPilot-restored recordings, respectively. Red boxes indicate zoomed insets. Temporal kymographs extracted along the white dashed lines are displayed beneath each main image. Scale bars represent 50 μm for the main images, 20 μm for the insets, and 40 μm vertically by 20 s horizontally for the kymographs.

h, Spatial map of neurons tracked across the four experimental paradigms. A total of 21 neurons are consistently identified and tracked across all stimulation conditions. Distinct marker shapes indicate the respective recordings, and colored masks denote the spatial footprints of individual neurons.

i-l, Calcium activity heatmaps of the 21 consistently tracked neurons across the four experimental paradigms: spinal column-fixed **i**, freely moving **j**, cold stimulation **k**, and nociceptive mechanical stimulation **l**. Grayscale bars indicate the $\Delta F/F$ signal amplitudes.

m, Quantification of neuronal activity statistics across the four experimental paradigms. Left: Distribution of the mean calcium activity for the tracked neurons. Each dot represents an individual neuron ($n = 21$). Right: Pairwise activity correlation within the neuronal population, illustrating consistent signal dynamics. Each dot represents the correlation coefficient for a distinct neuron pair. For the box plots, center lines denote the median, box limits represent the 25th and 75th percentiles, and whiskers extend to the extrema.

n, Spatial map of longitudinally tracked neurons in a spared nerve injury (SNI) mouse model of neuropathic pain. A total of 25 neurons are consistently identified and tracked across five imaging sessions (days 1, 5, 14, 21, and 28). Colored contours indicate individual spatial footprints, while distinct marker shapes denote the specific recording days. The background displays the standard deviation projection of the day 14 recording.

o-s, Calcium activity heatmaps of the 25 consistently tracked neurons across the five longitudinal recording sessions: day 1 **o**, day 5 **p**, day 14 **q**, day 21 **r**, and day 28 **s**. Grayscale bars indicate signal amplitudes.

882 **t**, Representative cross-day spatial displacement maps showing positional shifts from
883 day 1 (left) and day 21 (right), registered relative to the day 14 reference. Vector colors
884 and orientations visualize local displacement magnitude and direction.

885 **u**, Pairwise activity correlation matrices of the tracked neuronal population under four
886 experimental conditions (spinal column-fixed, freely moving, cold stimulation, and
887 nociceptive mechanical stimulation).

888 **v**, Pairwise activity correlation matrices of the same tracked neurons across longitudinal
889 sessions (days 1, 5, 14, 21, and 28).

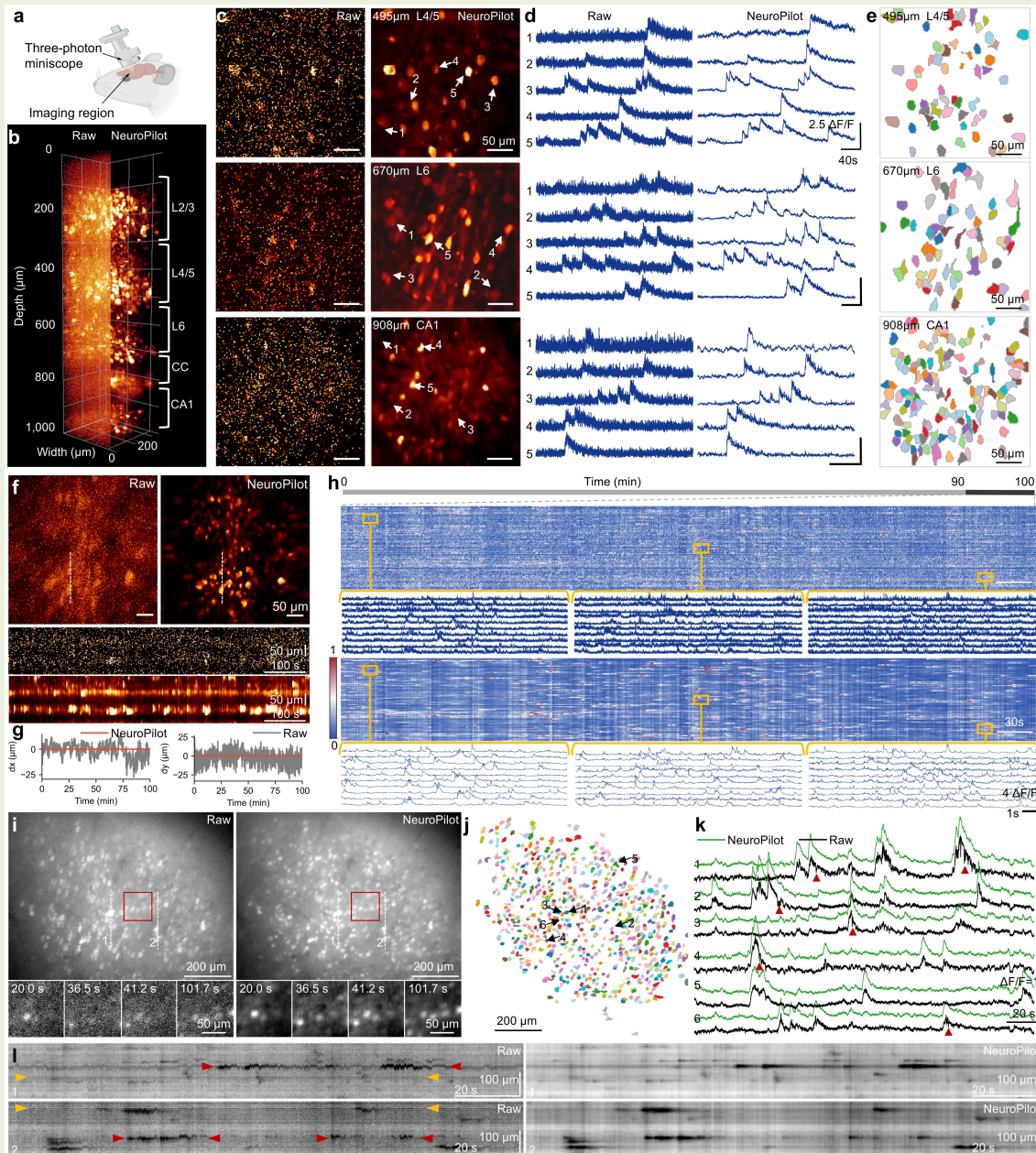


Fig. 4 | NeuroPilot generalizes across multi calcium imaging modalities.

a, Schematic of three-photon miniscope setup targeting deep brain regions.

b, Deep-tissue volumetric rendering of raw (left) and NeuroPilot-processed (right) three-photon data down to 1,000 μm , with brackets denoting cortical and hippocampal regions (L2/3, L4/5, L6, CC, and CA1).

c, Representative fields of view comparing raw (left) and NeuroPilot-processed (right) three-photon continuous recordings at depths of 495 μm (L4/5), 670 μm (L6), and 908 μm (CA1). Scale bars, 50 μm .

d, Calcium activity traces extracted from the representative neurons identified in **c** across the three depths (top: L4/5; middle: L6; bottom: CA1). Left, traces extracted from raw recordings; right, corresponding traces after NeuroPilot processing. Scale bars, 2.5 $\Delta F/F$ and 40 s.

e, Segmentation maps of neuronal spatial footprints detected after NeuroPilot restoration. Colored regions denote individual segmented cells. Scale bars, 50 μm .

f, Spatial and temporal restoration of a 100-minute three-photon recording in the **CA1** **TODO check which** region. Raw and NeuroPilot-processed MIPs are shown (top), alongside corresponding temporal kymographs (bottom) extracted along the dashed lines. Scale bars: 50 μm (top) and 50 μm by 100 s (bottom). **F panel** 把脑区标上

g, Temporal motion trajectories along the x- and y-axes over 100 minutes, comparing the raw uncorrected tissue displacement (gray) with the NeuroPilot-stabilized coordinates (red).

h, Population activity heatmaps and representative traces. Top panels show raw data, while bottom panels display the corresponding NeuroPilot-processed results over the 100-minute session. Expanded trace views (indicated by yellow boxes) highlight the robust removal of noise and preservation of calcium transients across different time points. Scale bars: 30 s (heatmaps), 1 s and 4 $\Delta F/F$ (traces).

i, Comparison of raw and NeuroPilot-processed one-photon miniscope imaging. Top panels show the full MIP fields of view, while bottom panels display magnified temporal snapshots of the region highlighted by the red box at 20.0, 36.5, 41.2, and 101.7 s. Scale bars, 200 μm (top) and 50 μm (bottom insets).

j, Segmentation map illustrating neuronal spatial footprints detected after NeuroPilot restoration. Uniquely colored masks define individual ROIs, with numbered arrows highlighting representative cells. Scale bar, 200 μm .

k, Calcium activity traces for the six representative neurons identified in **j**. Raw recordings (black) are plotted alongside corresponding NeuroPilot-processed signals (green). Red arrowheads highlight motion-induced artifacts in the raw traces that are absent in the processed signals. Scale bars, 1 $\Delta F/F$ and 20 s.

l, Kymographs comparing raw (left) and NeuroPilot-processed (right) signals along the dashed lines from **i**. Colored arrowheads indicate the removal of motion artifacts (red) and fixed-pattern noise (yellow). Scale bars, 100 μm and 20 s.

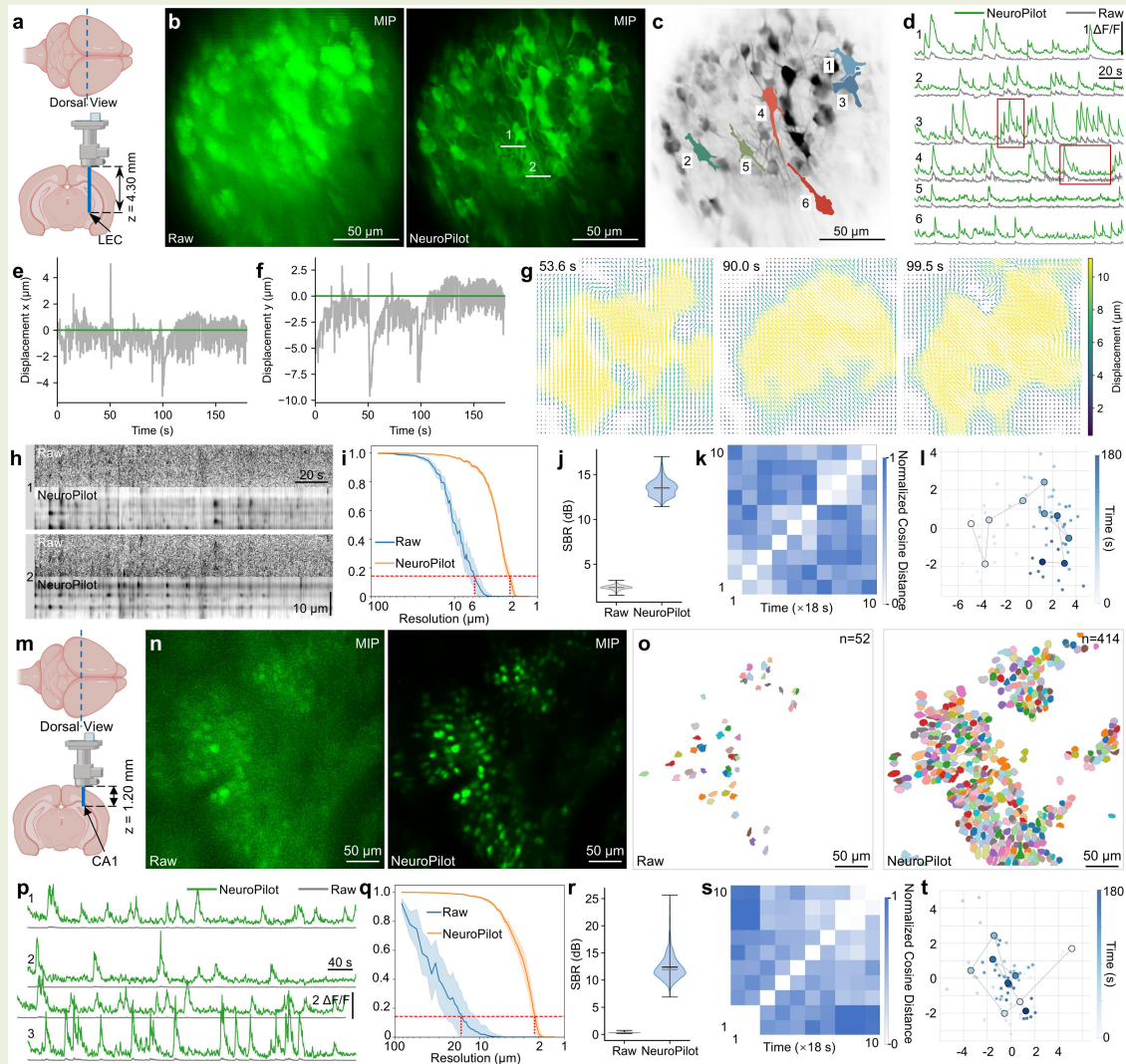


Fig. 5 | NeuroPilot facilitates high-fidelity calcium imaging in deep brain regions.

a, Deep-brain two-photon miniscope imaging setup targeting the LEC at a depth of $z = 4.30$ mm. The coronal section (bottom) illustrates the lens implantation trajectory corresponding to the dashed line in the dorsal view (top).

b, Comparison of raw (left) and NeuroPilot-processed (right) MIP images from the LEC. Numbered white lines denote cross-sections for kymograph analysis. Scale bars, $50\ \mu\text{m}$.

c, Segmented neuronal spatial footprints (colored masks) in the LEC after NeuroPilot restoration.

d, Comparative calcium activity traces from neurons 1 – 6 marked in **c**. Raw recordings (gray) are plotted alongside NeuroPilot-processed signals (green). Red boxes denote recovered calcium transients. Scale bars, $1\ \Delta F/F$ and $20\ \text{s}$.

941 **e-f**, Temporal motion displacement trajectories along the x- and y-axes. The plots
 942 compare raw uncorrected drift (gray) with NeuroPilot-stabilized coordinates (green).
 943 **g**, Spatial displacement fields at representative time points. Arrows and colors denote
 944 the direction and magnitude (μm) of local motion, respectively.
 945 **h**, Spatiotemporal kymographs corresponding to lines 1 and 2 in **b**. Comparison
 946 between raw and NeuroPilot-processed data highlights the removal of shot noise and
 947 motion-induced jitter, revealing distinct calcium transients across time. Scale bars, 20 s
 948 and 10 μm .
 949 **i**, Spatial resolution comparison using FRC analysis. Solid lines and error bands (mean
 950 $\pm$ s.d.) correspond to raw and NeuroPilot-processed recordings. Vertical red dotted lines
 951 denote the estimated resolution values at the 1/7 threshold (horizontal dashed line).
 952 Resolution is enhanced from $\sim 6 \mu\text{m}$ (raw) to $\sim 2 \mu\text{m}$ (NeuroPilot).
 953 **j**, Quantitative comparison of SBR for raw and NeuroPilot-processed recordings.
 954 **k**, Temporal similarity matrix of population activity. Each entry represents the
 955 normalized cosine distance between activity vectors across 18-s time bins. The 10×10
 956 matrix spans the total recording duration (10×18 s).
 957 **l**, Two-dimensional visualization of neural population trajectories. Points are color-
 958 coded by time (0 – 180 s) to illustrate the temporal progression of activity states.
 959 **m**, Schematic of the imaging setup targeting the CA1 region at a depth of $z = 1.20$ mm.
 960 **n**, Comparison of raw (left) and NeuroPilot-processed (right) MIP images from the
 961 hippocampal CA1. Scale bars, 50 μm .
 962 **o**, Segmentation maps comparing raw (left, $n = 52$) and NeuroPilot-restored (right, $n =$
 963 414) spatial footprints in CA1. Scale bar, 50 μm .
 964 **p**, Example calcium activity traces from hippocampal CA1 neurons. Raw recordings
 965 (gray) are plotted alongside NeuroPilot-processed signals (green). Scale bars, 2 $\Delta F/F$
 966 and 40 s.
 967 **q**, FRC curves comparing the spatial resolution of raw (blue) and NeuroPilot-restored
 968 (orange) CA1 images. Shaded regions indicate s.d. The red dashed line marks the 1/7
 969 threshold, illustrating a shift in estimated resolution from $\sim 18 \mu\text{m}$ to $\sim 2 \mu\text{m}$.
 970 **r**, SBR comparison for raw and NeuroPilot-restored neurons in CA1.
 971 **s**, Population activity similarity matrix (18-s bins) based on normalized cosine distance.

972 t, Low-dimensional projection of population trajectories color-coded by session time (0-
973 180 s).
974

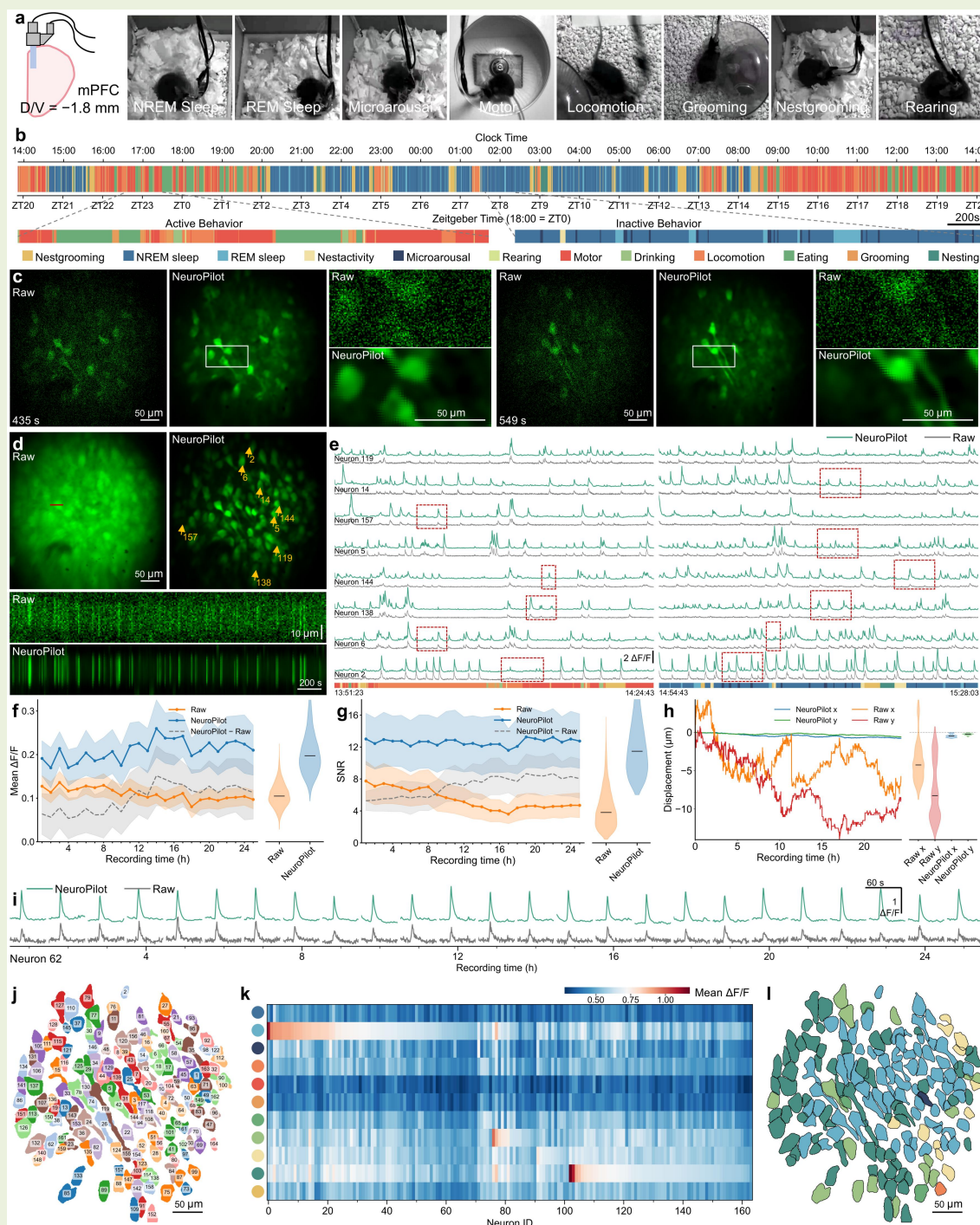


Fig. 6 | NeuroPilot underpins 24-hour imaging of mPFC neuronal ensembles over a circadian cycle.

a, Schematic of the mPFC imaging site ($D/V = -1.8$ mm) and example video frames capturing representative behavioral states annotated over the 24-hour period.

b, Comprehensive behavioral ethogram spanning the continuous 24-hour recording. The temporal distribution of 12 behaviors is mapped against Clock and Zeitgeber time

(18:00 = ZT0). Lower panels display magnified views of characteristic active and inactive periods. Scale bar, 200 s.

c, Comparison of raw and NeuroPilot-enhanced calcium imaging data extracted at 435 s and 549 s. White boxes indicate regions shown at higher magnification in the right panels. Scale bars, 50 μm .

d, STD projections (top) and kymographs (bottom) comparing raw and NeuroPilot recordings. Kymographs show temporal dynamics along the red cross-section. Scale bars: 50 μm (top), 10 μm and 200 s (bottom).

e, Comparison of raw (gray) and NeuroPilot-processed (green) temporal dynamics for neurons identified in **d**. Red dashed boxes highlight improvements in SNR and transient detectability. Color-coded bars at the bottom indicate the corresponding behavioral states during the recording segments. Scale bar, 2 $\Delta F/F$.

f-g, NeuroPilot enhances and stabilizes calcium signal amplitude **f** and SNR **g** over long-term recordings. Mean $\Delta F/F$ and SNR of mPFC population activity are plotted as a function of recording time for raw (orange) and NeuroPilot-processed (blue) data. Dashed gray traces denote the performance gain. Shaded areas represent $\pm$ s.e.m. across neurons.

h, Cumulative lateral motion displacement along the x- (solid) and y-axes (dashed) over 24 hours. Raw recordings (orange) exhibit severe progressive drift, which is successfully stabilized by NeuroPilot (blue).

i, Preservation of spike timing consistency over long-term recordings. Scale bars, 60 s and 1 $\Delta F/F$.

j, Spatial footprints of the segmented mPFC neuronal ensemble. Individual neurons ($n = 164$) are delineated by distinct colors and annotated with unique numerical identifiers. Scale bar, 50 μm .

k, Heatmap of mean $\Delta F/F$ for all identified neurons across different behavioral states. Rows are color-coded to match the behaviors in **b**. Neurons (x-axis) are sorted by their activity profiles to reveal behavioral tuning.

l, Anatomical projection of the functional clusters from **k**. Neuronal footprints are color-coded by their preferred behavioral state. Scale bar, 50 μm .

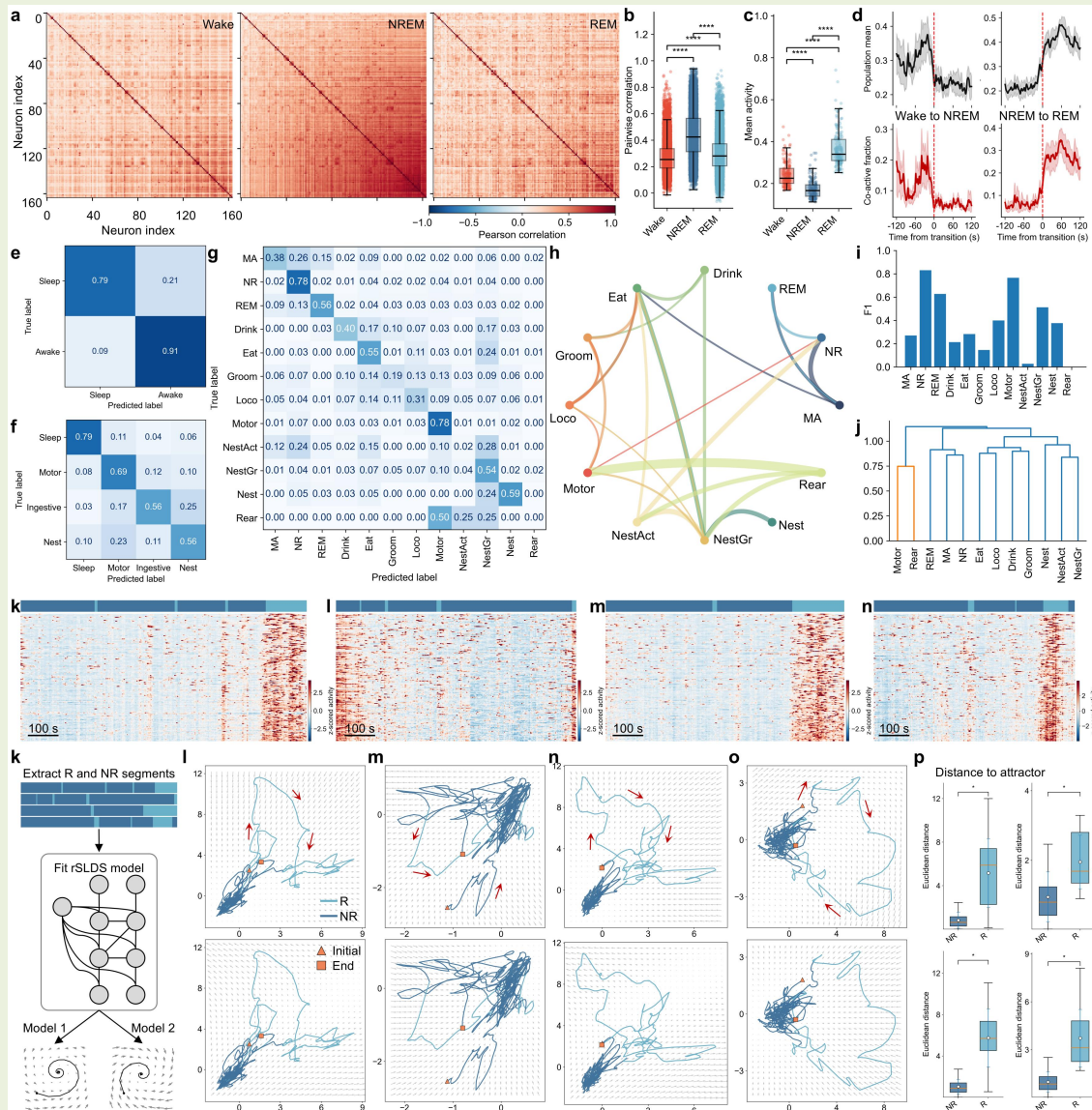


Fig. 7 | Decoding 24-Hour mPFC population dynamics across the behavioral spectrum.

a, Pairwise activity correlation matrices of mPFC neurons during Wake, NREM, and REM states, revealing state-dependent population structure.

b, Distribution of pairwise correlations across behavioral states, showing increased population synchrony during NREM and REM relative to wakefulness.

c, Mean neuronal activity across states, highlighting state-dependent modulation of population activity levels.

d, Transition-aligned population dynamics. Top, population mean activity across Wake-to-NREM and NREM-to-REM transitions. Bottom, corresponding co-active neuron fraction, showing coordinated changes in population synchrony across state transitions.

1023 **e**, Binary sleep-wake decoding.

1024 **f**, Decoding of higher-order behaviors (sleep, motor, ingestive, and nest).

1025 **g**, Fine-grained decoding of the 24-hour ethogram. All matrices in **e-g** display row-
 1026 normalized predictions from a multinomial logistic regression model trained on mPFC
 1027 population activity and evaluated on an independent test set.

1028 **h**, Representational overlap across the behavioral spectrum. Chord diagram mapping the
 1029 prominent decoding cross-talk between states. Nodes represent individual behaviors,
 1030 and chord thickness is proportional to the off-diagonal misclassification probabilities,
 1031 indicating shared neural representations.

1032 **i**, State-specific decoding efficacy. F1 scores evaluating the classification performance
 1033 for each individual behavioral category.

1034 **j**, Hierarchical organization of behavioral representations. Dendrogram clustering the 12
 1035 behavioral states based on the pairwise similarity of their mPFC population activity
 1036 patterns.

1037 **k-n**, State-specific population dynamics across NREM-REM sleep transitions. Z-scored
 1038 activity heatmaps of 164 mPFC neurons capturing continuous transitions between
 1039 NREM and REM sleep. Aligned behavioral hypnograms are shown above. Scale bars,
 1040 100 s.

1041 **k**, Inference of latent neural dynamics using rSLDS. Computational pipeline showing
 1042 the extraction of REM/NREM segments to fit an rSLDS model, which decomposes the
 1043 sleep-state neural dynamics into distinct continuous vector fields (Model 1 and 2).

1044 **l-o**, Phase space trajectories and underlying vector fields during sleep. rSLDS-inferred
 1045 latent trajectories for REM (light blue) and NREM (dark blue) sleep overlaid on local
 1046 vector fields (gray arrows). Triangles and squares indicate temporal start and end points,
 1047 respectively. Red arrows (top) indicate non-linear trajectory deviations from local flow
 1048 symmetry.

1049 **p**, Quantitative divergence of neural trajectories from underlying attractors. Box plots
 1050 comparing the Euclidean distance between instantaneous mPFC latent states and their
 1051 corresponding rSLDS-inferred attractors during NREM (NR) versus REM (R) sleep.
 1052 White circles denote means; boxes, interquartile ranges; whiskers, data range. $*P < 0.05$,
 1053 Wilcoxon rank-sum test.